

Skills Summary

Equations with Brackets and Fractions

Use this reference while completing your worksheet or when studying.

Skill 1 — Distributing over a bracket

Rule: $a(b + c) = ab + ac$ — the outside factor multiplies *every* term inside, not just the first.

A minus sign outside counts as part of the factor, so it changes the sign of every term inside. After distributing, the equation is an ordinary two-step equation.

Example: Solve $4(x + 3) = 32$.

Step 1. A bracket blocks any collecting, so the first job is to distribute the outside factor across both terms inside.

Step 2. That gives a two-step equation: subtract the constant from both sides, then divide both sides by the coefficient.

$$\Rightarrow x = 5$$

Try it: Solve $6(x - 2) = 30$.

check: $x = 5$

Skill 2 — Clearing a denominator

Rule: multiply *both whole sides* by the least common denominator. Every term is multiplied — including terms that carry no fraction at all.

That single move turns a fraction equation into an ordinary one, which is why it is worth doing before anything else.

Example: Solve $\frac{x}{5} + 2 = 6$.

Step 1. A denominator is in the way, so multiply both sides by it and the fraction disappears in one step.

Step 2. Multiplying every term by five clears the fraction and leaves a two-step equation; subtract the constant, then divide.

$$\Rightarrow x = 20$$

Try it: Solve $\frac{3x}{4} = 9$.

check: $x = 12$

(!) Watch out: the two classic slips are multiplying only the term that had the fraction, and distributing to only the first term inside a bracket. In both cases something on the page was left untouched — check that every term took the move.

Skill 3 — When a bracket sits over a denominator

Method: a fraction bar groups everything above it, so treat the numerator as a bracket. Multiply both sides by the least common denominator first; the numerators are then multiplied as whole brackets, and only afterwards are they distributed.

Example: Solve $\frac{x-1}{3} = \frac{x+5}{6}$.

Step 1. Both sides are single fractions with grouped numerators, so clear both denominators at once with their least common multiple.

Step 2. Multiplying through leaves a bracket on the left to distribute; then collect the variable terms on one side and finish.

$$\Rightarrow x = 7$$

Try it: Solve $\frac{x+2}{2} = \frac{x+8}{4}$.

check: $x = 7$